Astrodynamics II: Midterm Exam

October 23, 2012 3:00-4:30PM (90 minutes)

1. (25) Consider a coplanar transfer between circular orbits using one tangential burn and one non-tangential burn. As is shown in the left hand side of Figure below, the first tangential burn occurs at apoapsis and the second non-tangential burn occurs at the designated true anomaly $\nu_{trans,b}$ of the elliptic transfer orbit.

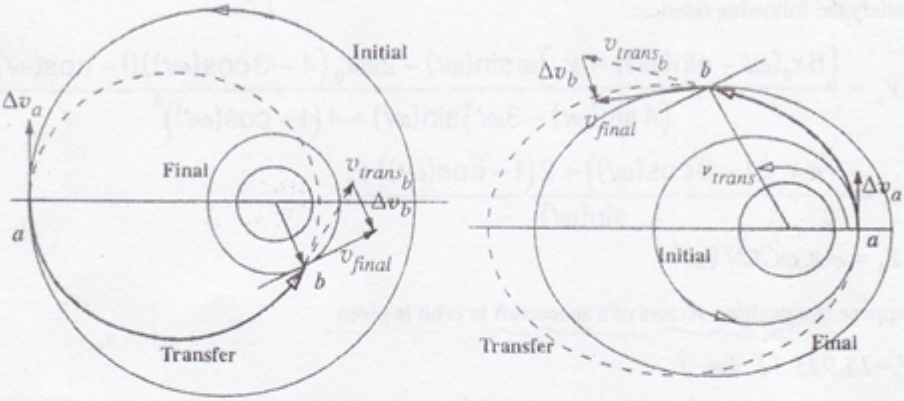

Figure 6-7. **One-Tangent Burns.** For both circular (left) and elliptical (right) orbit changes, the true anomalies of the transfer and final orbits are equal. The maneuver starts at apoapsis (left, $\nu > 180°$) and periapsis (right, $\nu < 180°$) of the transfer orbit.

$\tan\phi = \frac{e\sin\nu}{1+e\cos\nu}$ $E = \frac{e+\cos\nu}{1+e\cos\nu}$

Given $r_{initial}$, r_{final}, $\nu_{trans,b}$, describe how to determine e_{trans}, a_{trans}, ΔV_a, ΔV_b, τ_{trans}. Assume that the flight path angle $\varphi_{trans,b}$ and the Eccentric anomaly $E_{trans,b}$ are known.

2. (25) Consider a satellite initially on a geosynchronous circular equatorial orbit. After performing a maneuver, it is now located at the following relative position and velocity with respect to its non-maneuvering position in the local RSW coordinates:

$$(x, y, z) = (-110, 70, 10)km$$
$$(\dot{x}, \dot{y}, \dot{z}) = (-10, 20, 1)m/\sec$$

If the satellite is to initiate a two-impulse maneuver to return to its original non-maneuvering location in 2 hours, what are the changes in velocities at the initial and terminal time, i.e., ΔV_1 and ΔV_2?

Hint: The solution to the Hill-Clohessy-Wiltshire (HCW) equation is given as

$$x(t)=\frac{\dot{x}_0}{\omega}\sin(\omega t)-\left(3x_0+\frac{2\dot{y}_0}{\omega}\right)\cos(\omega t)+\left(4x_0+\frac{2\dot{y}_0}{\omega}\right)$$

$$y(t)=\left(6x_0+\frac{4\dot{y}_0}{\omega}\right)\sin(\omega t)+\frac{2\dot{x}_0}{\omega}\cos(\omega t)-\left(6\omega x_0+3\dot{y}_0\right)t+\left(y_0-\frac{2\dot{x}_0}{\omega}\right)$$

$$z(t)=\frac{\dot{z}_0}{\omega}\sin(\omega t)+z_0\cos(\omega t)$$

If the relative position in the HCW equation is equal to zero after time t, the initial position and velocity must satisfy the following relation:

$$\dot{y}_0=\frac{\left[6x_0(\omega t-\sin(\omega t))-y_0\right]\omega\sin(\omega t)-2\omega x_0\left(4-3\cos(\omega t)\right)\left(1-\cos(\omega t)\right)}{\left(4\sin(\omega t)-3\omega t\right)\sin(\omega t)+4\left(1-\cos(\omega t)\right)^2}$$

$$\dot{x}_0=-\frac{\omega x_0\left(4-3\cos(\omega t)\right)+2\left(1-\cos(\omega t)\right)\dot{y}_0}{\sin(\omega t)}$$

$$\dot{z}_0=-z_0\omega COT(\omega t)$$

3. (50) Suppose two position vectors of a spacecraft in orbit is given

$\vec{r}_0=15{,}945.34\ km\ \hat{I}$

$\vec{r}=12{,}214.84\ km\ \hat{I}+10{,}249.47\ km\ \hat{J}$

I. (5) State the definition of general Lambert problem precisely.

II. (5) Explain how to solve Lambert original formulation geometrically. Draw a schematic diagram of how to locate foci.

III. (5) What is the Lambert minimum-energy solution? Draw a schematic diagram showing the relation between initial radius, terminal radius, chord length, locations of foci, etc.

IV. (15) For the given position vectors, find the associated velocity vectors $(\vec{v}_0,\vec{v})$ for the short way transfer by applying Lambert minimum-energy solution.

V. (20) Use $(\vec{r}_0,\vec{v}_0)$ to determine the associated orbital elements.

Hint: Use the following information if necessary:

$$p_{min}=\frac{rr_0}{c}(1-\cos\Delta v)$$

$$f=1-\frac{r}{p}[1-\cos\Delta v],\qquad g=\frac{rr_0\sin\Delta v}{\sqrt{\mu p}}$$

$$\dot{f}=\sqrt{\frac{\mu}{p}}\tan\left(\frac{\Delta v}{2}\right)\left[\frac{1-\cos(\Delta v)}{p}-\frac{1}{r}-\frac{1}{r_0}\right],\quad \dot{g}=1-\frac{r_0}{p}(1-\cos(\Delta v))$$

$$\vec{e}=\frac{\left(v^2-\frac{\mu}{r}\right)\vec{r}-(\vec{r}\cdot\vec{v})\vec{v}}{\mu},\qquad \mu=398{,}600.4418\,km^3/sec^2$$